DEVOPS INTRODUCTION

DIGITAL TRANSFORMATION

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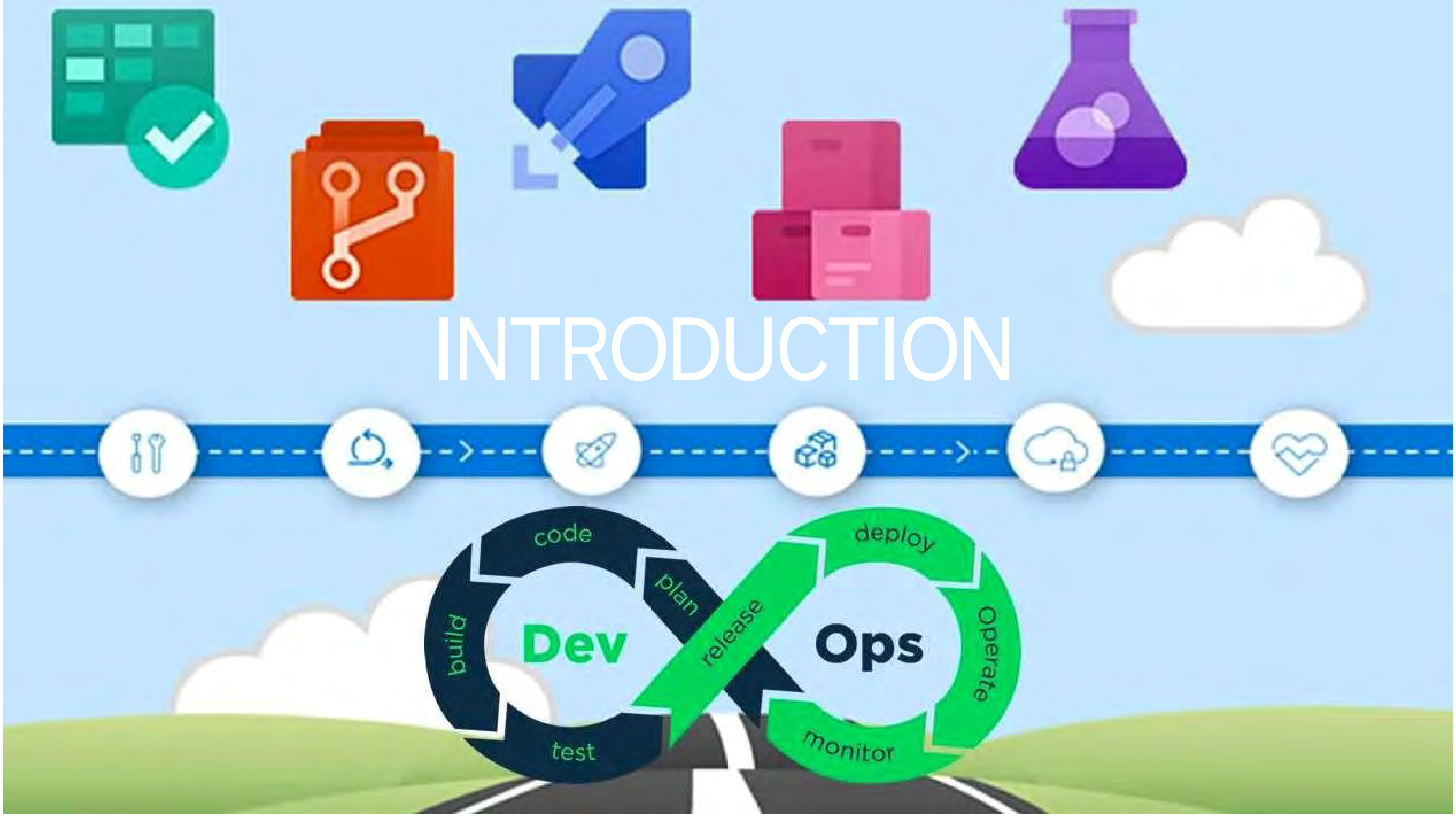
INTRODUCTION

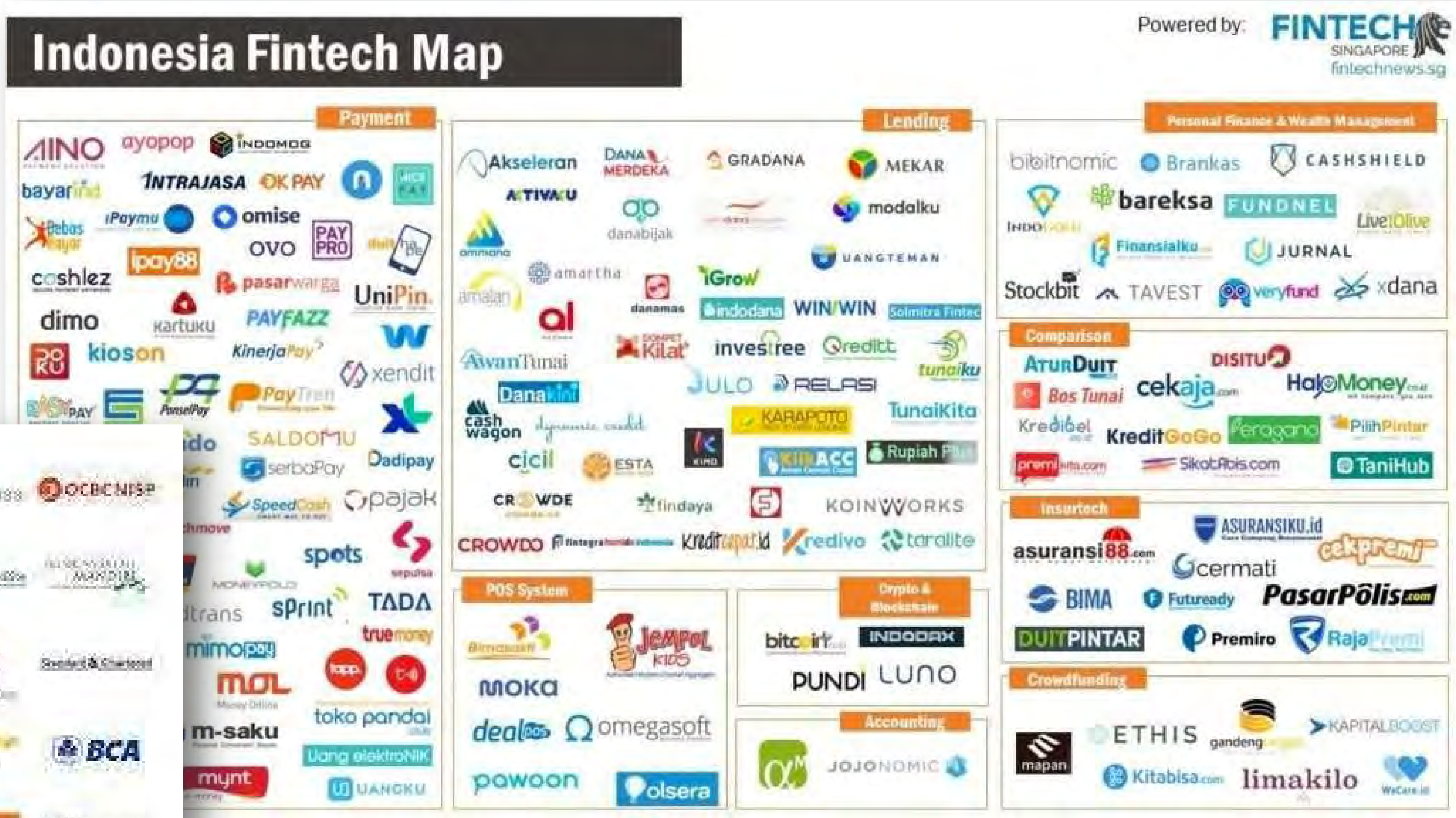


Agenda

1. Digital Transformation
2. Introduction of DevOps
3. Technology Component
4. Learning Path
5. Adoption
6. Maturity Model
7. How to start

Financial Industry Transformation

Boost Business with Digital Transformation




Boost Business with Digital Transformation
Business Agility is a must
The Fact of the BusinessNow:
• Kebutuhan Businesssangat dinamis dan cepat di EraDigitalisasi
• Tidak mudah memprediksi Trenddan Kebutuhan dimasa yang akandatang
• Mayoritas kebutuhan Businesssangat bergantung pada Teknologi


Business Agility Becoming Challenge of IT
IT Harus memiliki strategy
untuk mengimbangi kebutuhan business
yang sangat cepat dan dinamis
Digital Transformation is a MUST

Digital Transformation

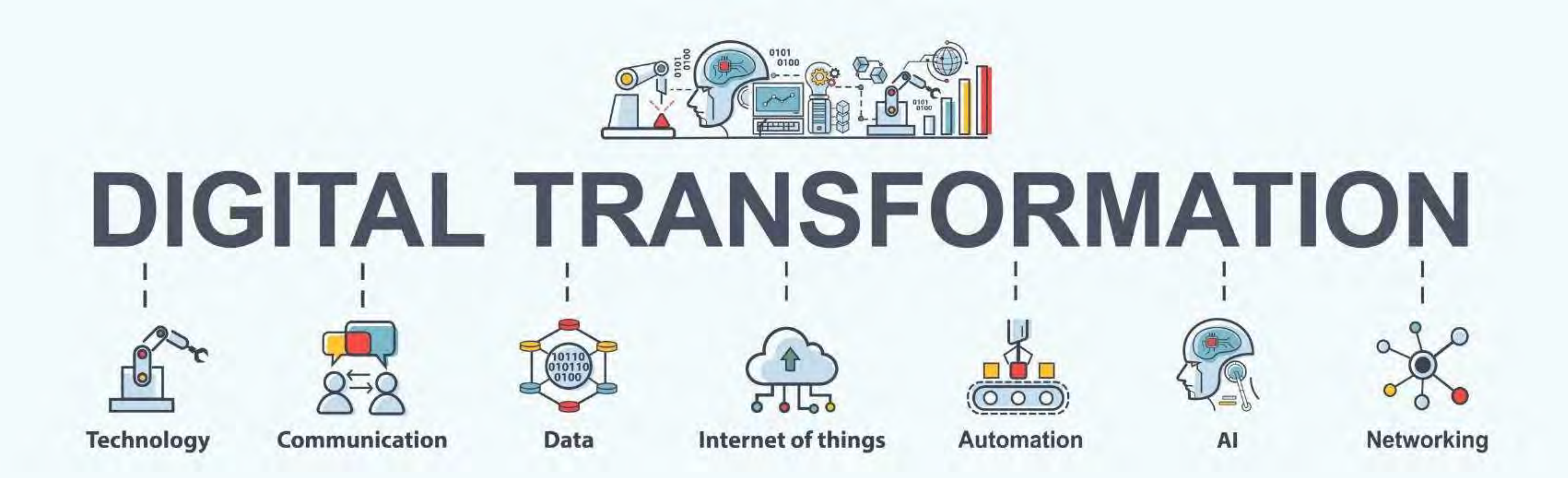
Business Architecture

InformationArchitecture

Application Architecture

TechnologyArchitecture

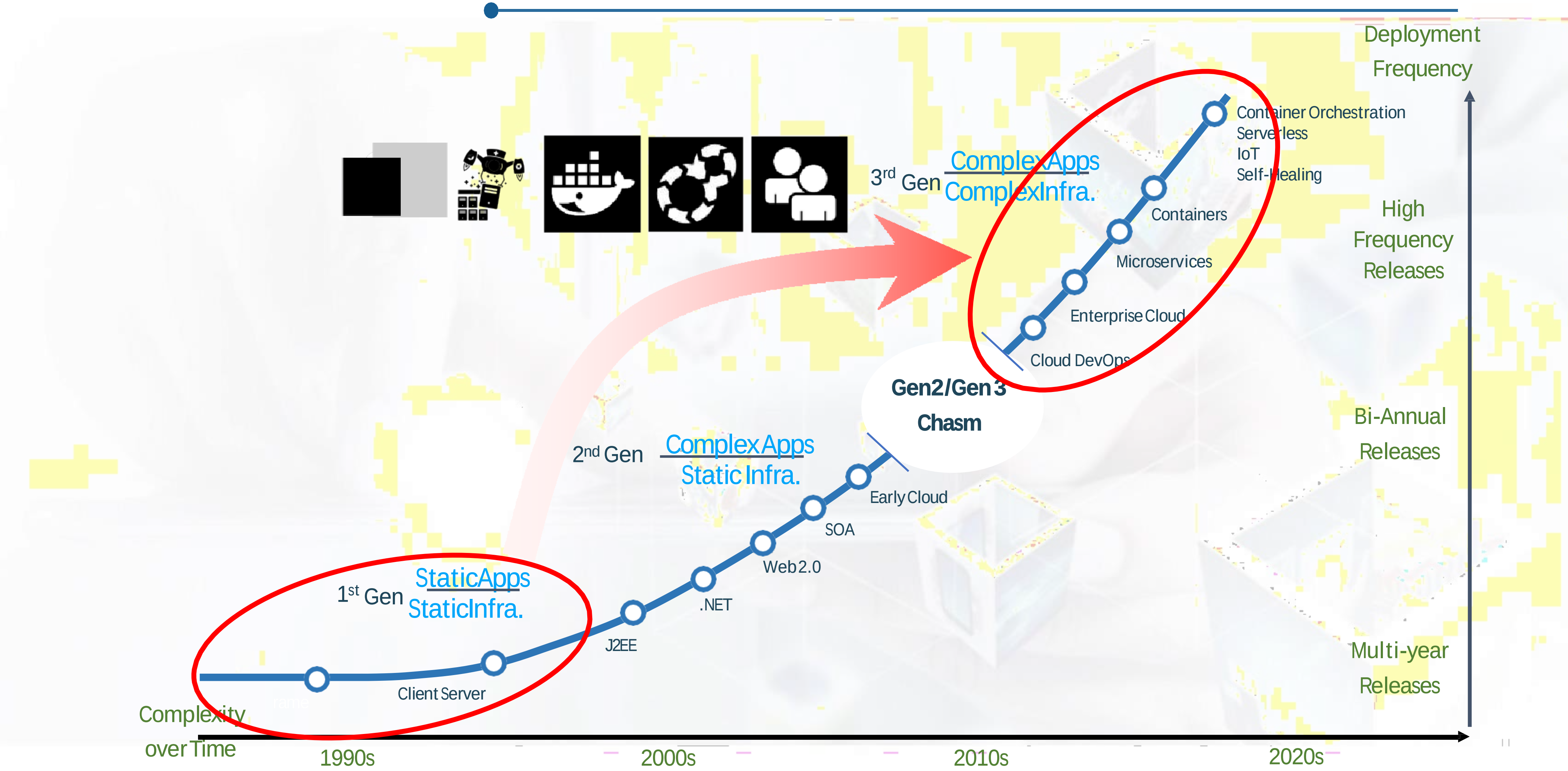
Every organization - whether services, financial, manufacturing or healthcare is becoming a technology company



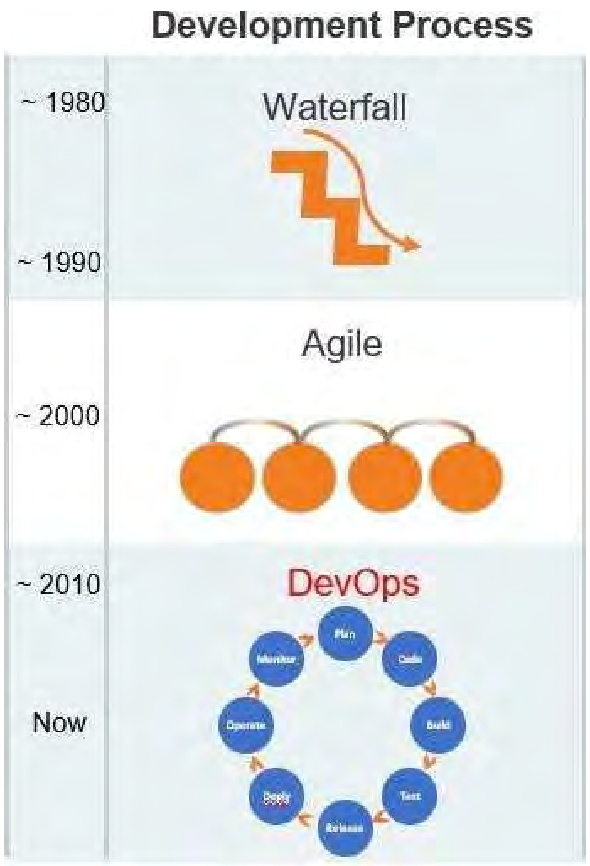
Technology Architecture

Every organization - whether services, financial, manufacturing or healthcare is becoming a technology company

Digital Transformation

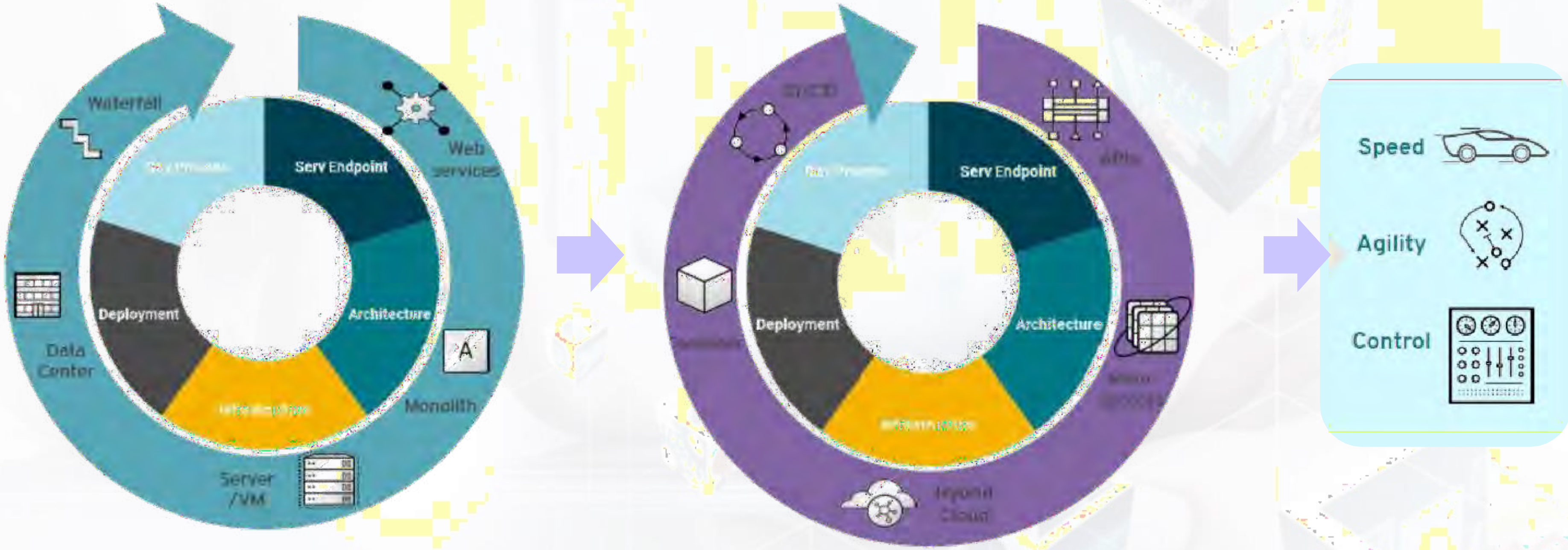


Digital Transformation - 1. Development Process



Digital Transformation - Technology Architecture

Changing enterprises and entire industries in fundamental ways

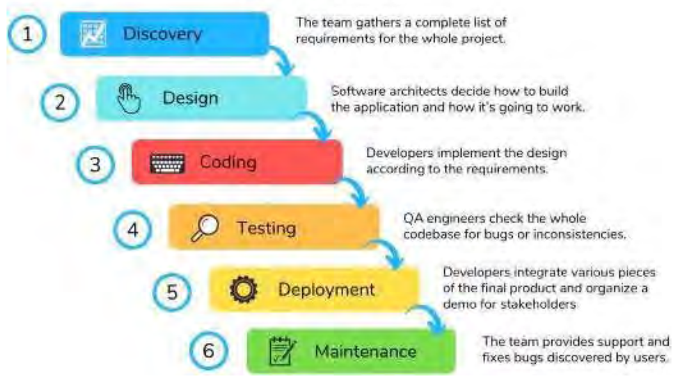


Every organization - whether services, financial, manufacturing or healthcare is becoming a technology company

SDLC (Software Development Life Cycle) & Operation Methodology

IT Development

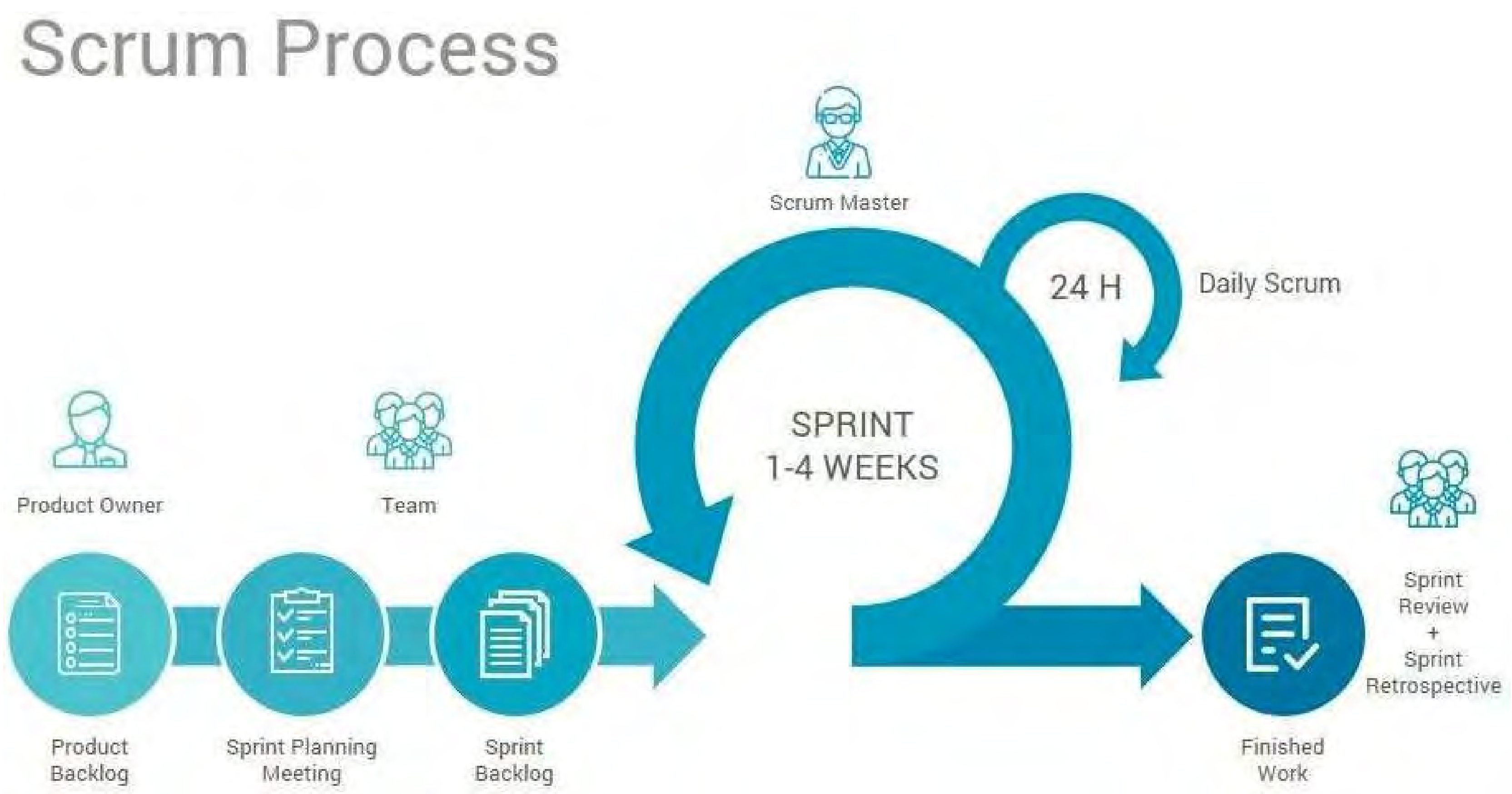
Waterfall Methodology



Agile Methodology

IT Operation

SDLC Agile - SCRUM



Why Agile +DevOps?

Water FallMethodology

Process within 6Months

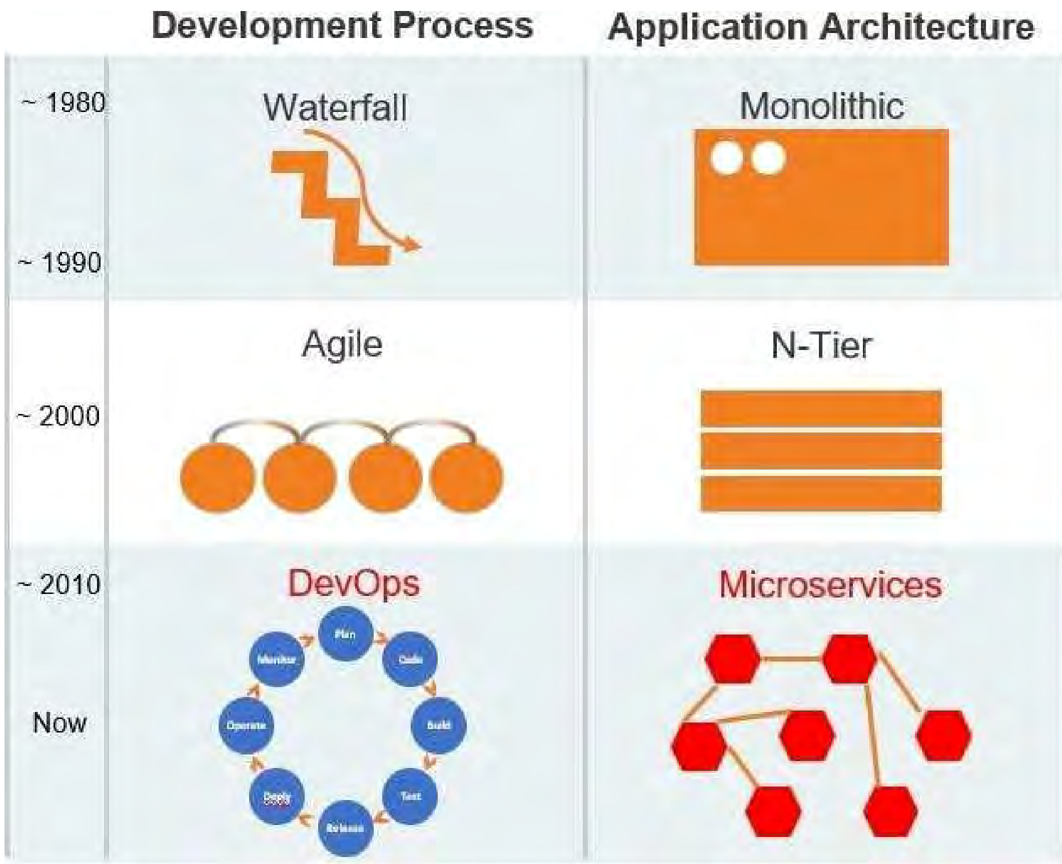
Agile Methodology + DevOps

Process within 3 x 1 Months

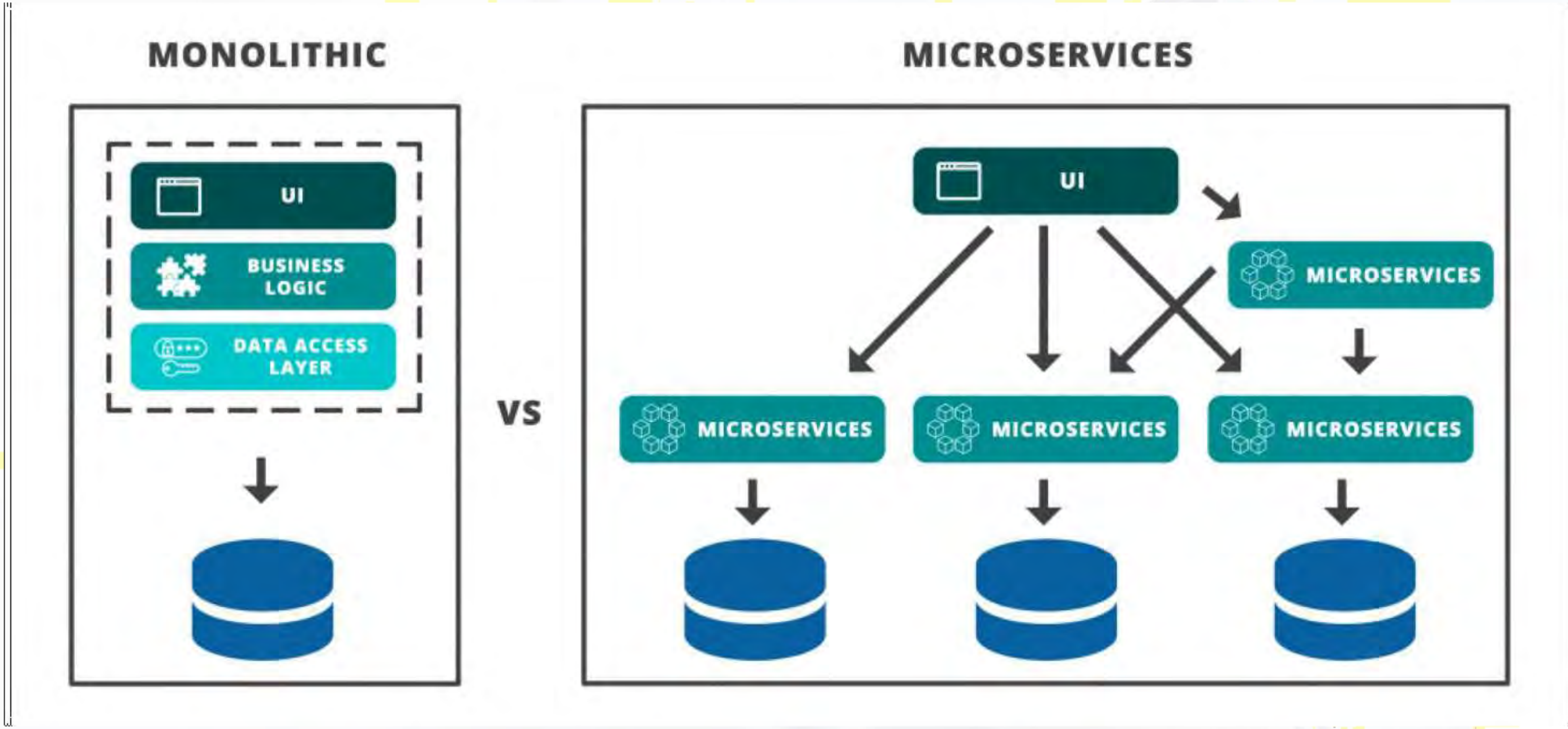
Automation

- *Continuous Integration*
- *Continuous Delivery*

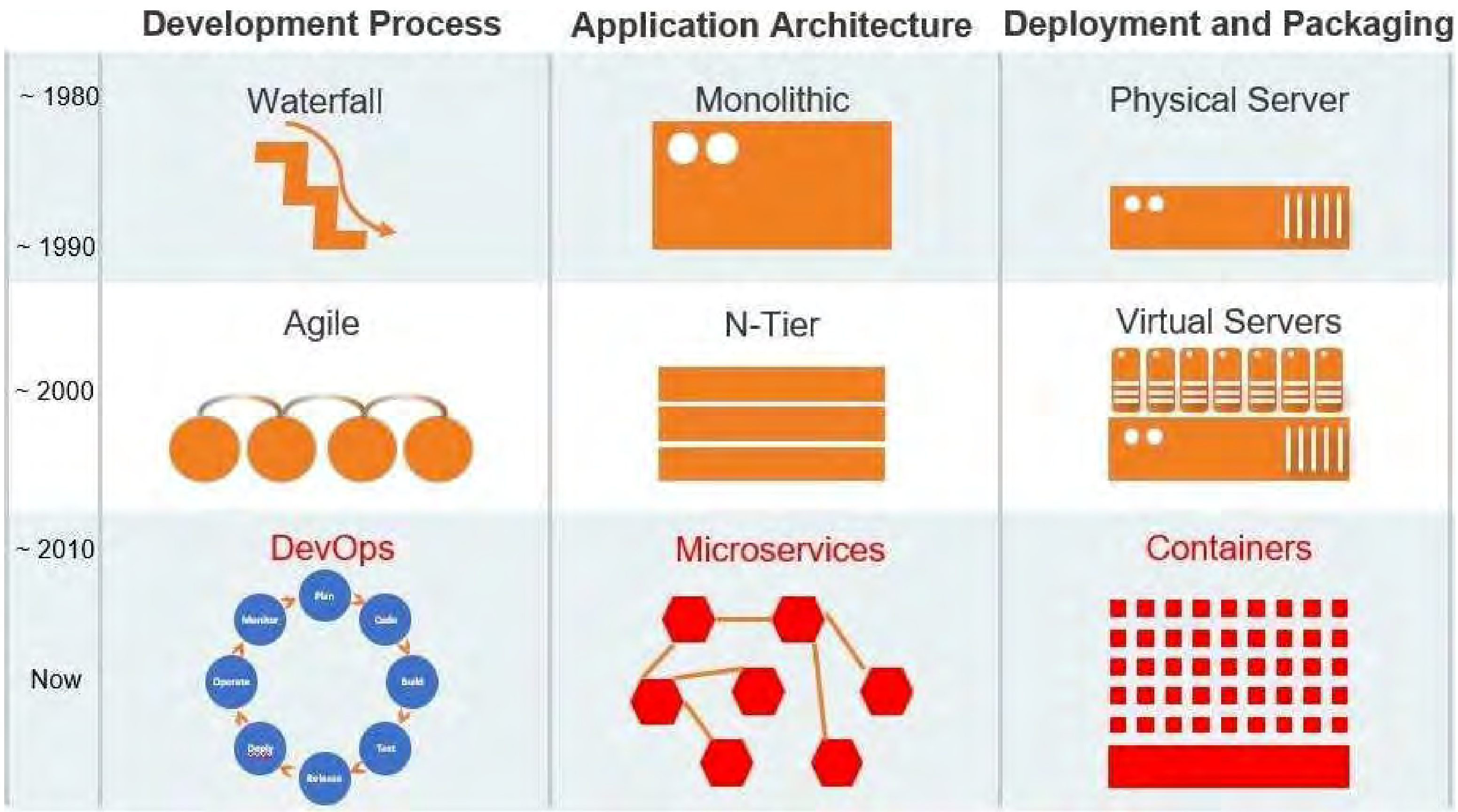
Digital Transformation – 2. Application Architecture



Application Architecture



Digital Transformation – 3. Deployment and Packaging



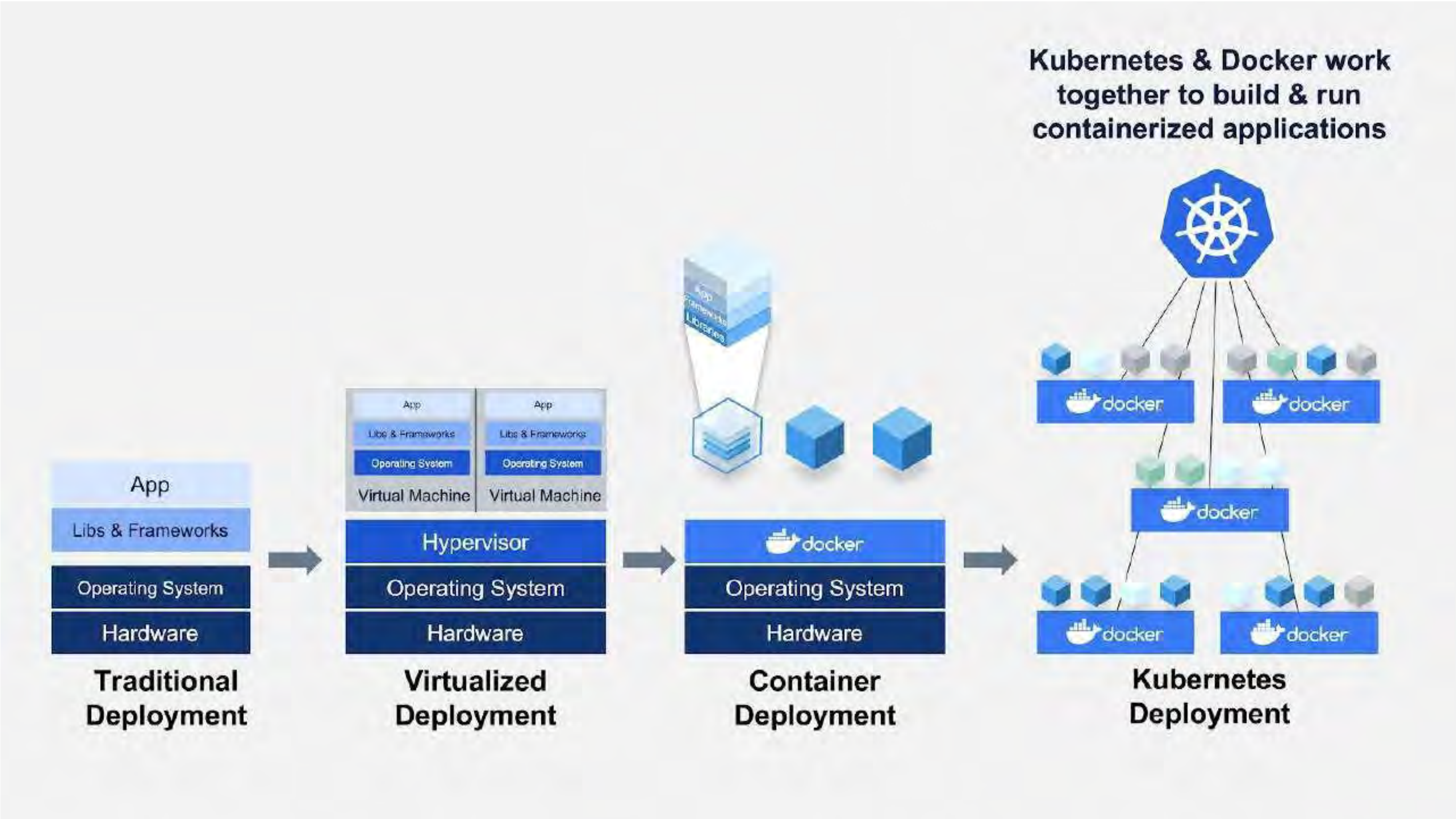
Container

What are Containers?

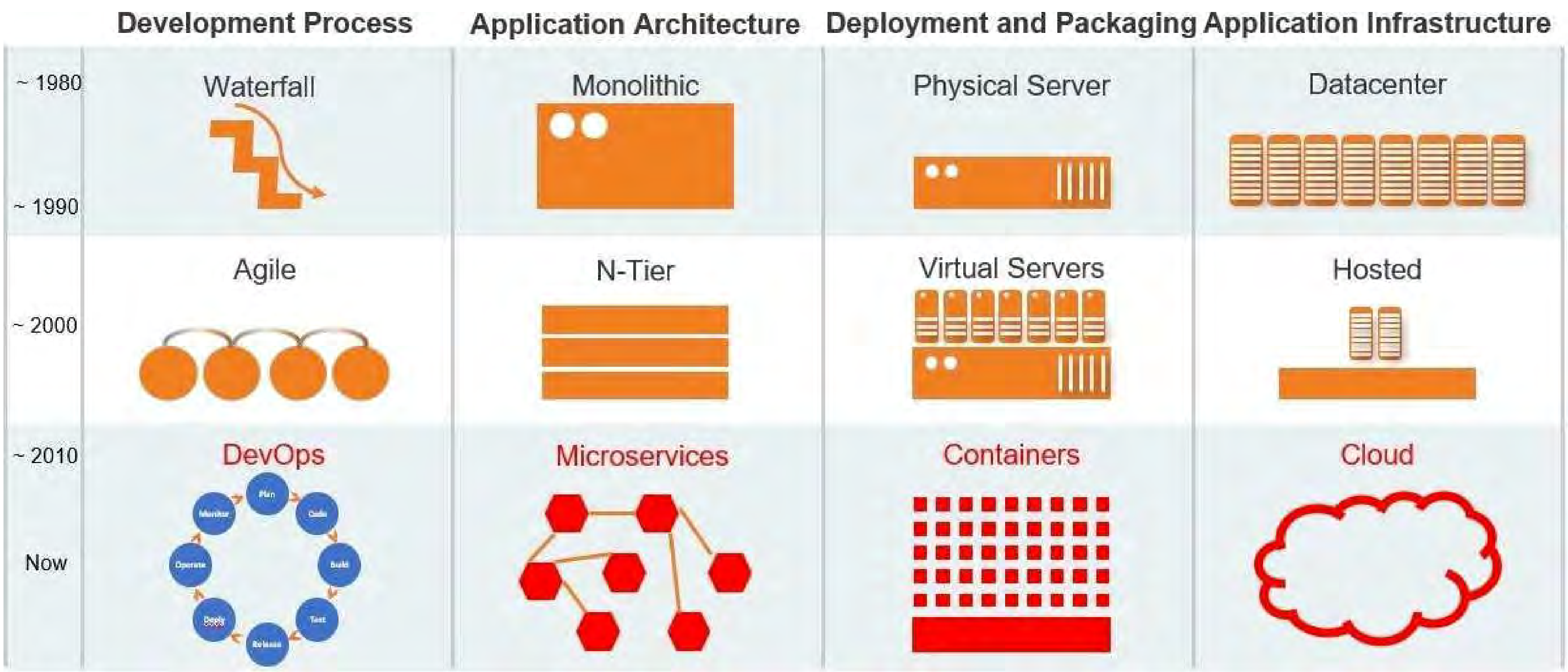
- Simpler, lighter, and denser than virtual machines
- Portable across different environment
- enable CI/CD

- Package my application and all of its dependencies
- Deploy to any environment in seconds
- Easily access and share containerized component

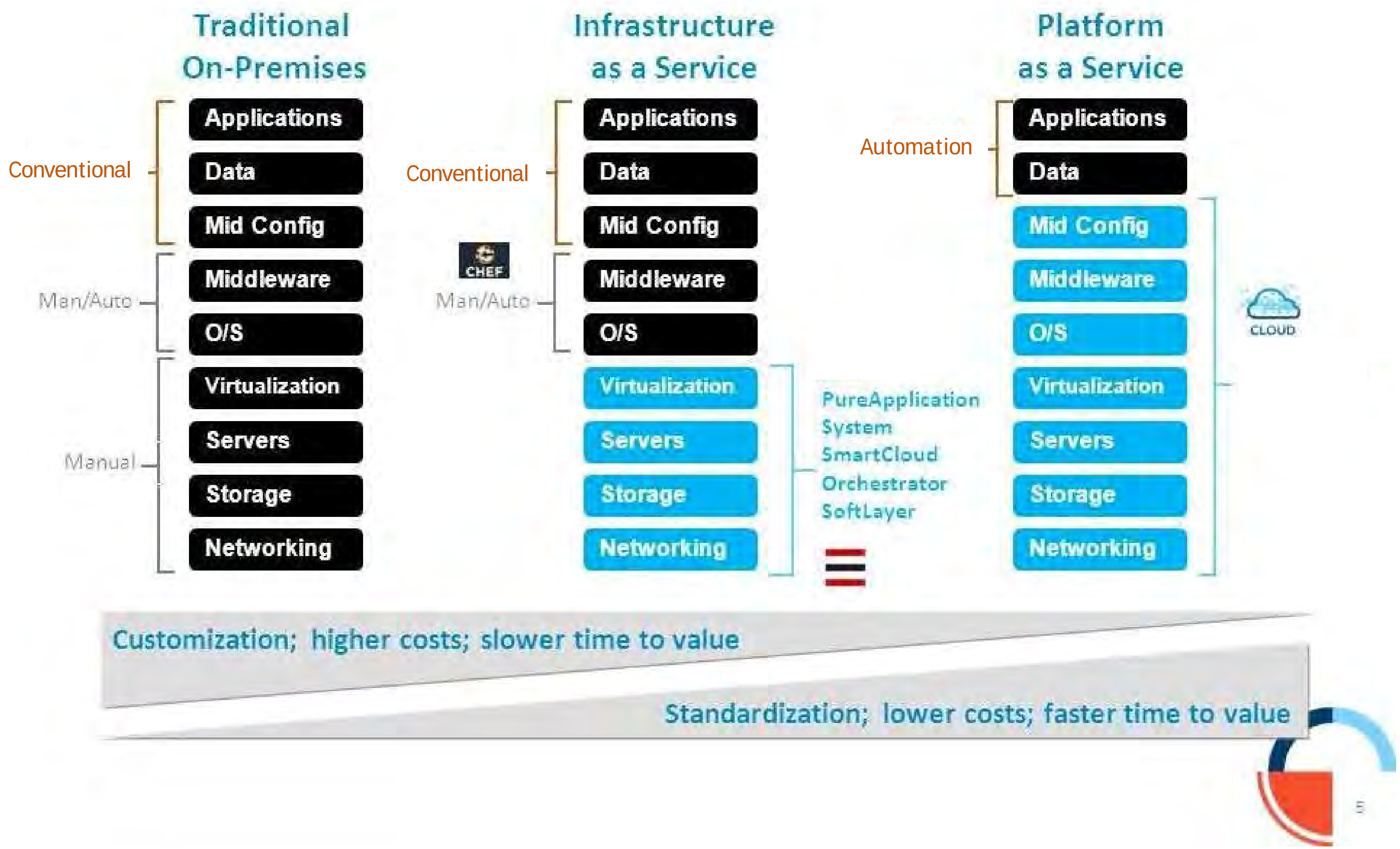
Technology Evolution



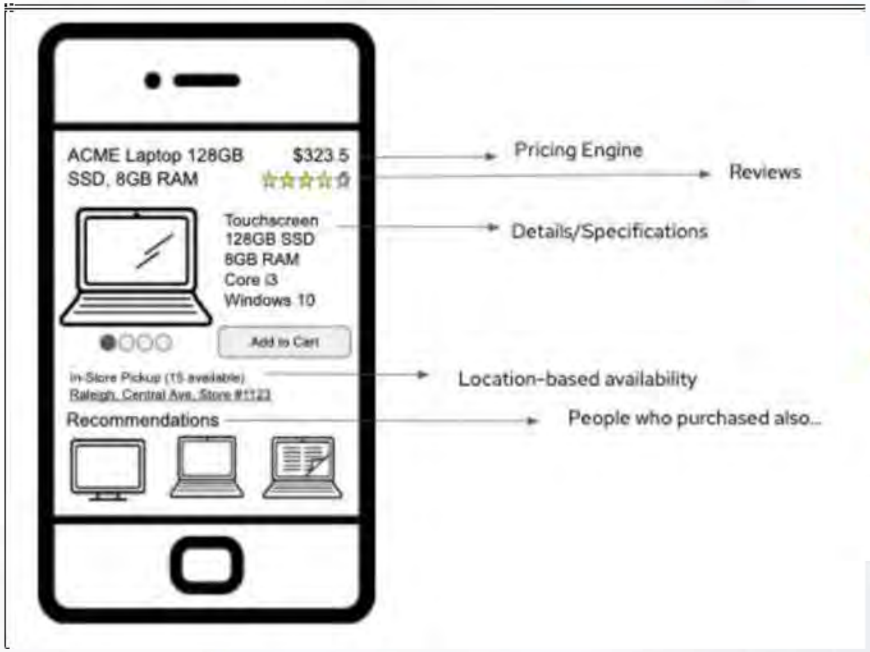
Digital Transformation – 4. Application Infrastructure



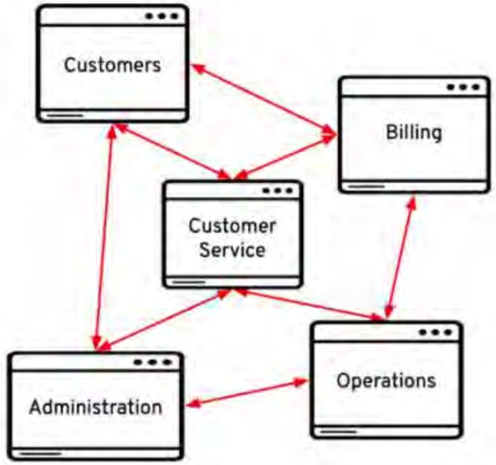
Technology Platform Supporting Devops



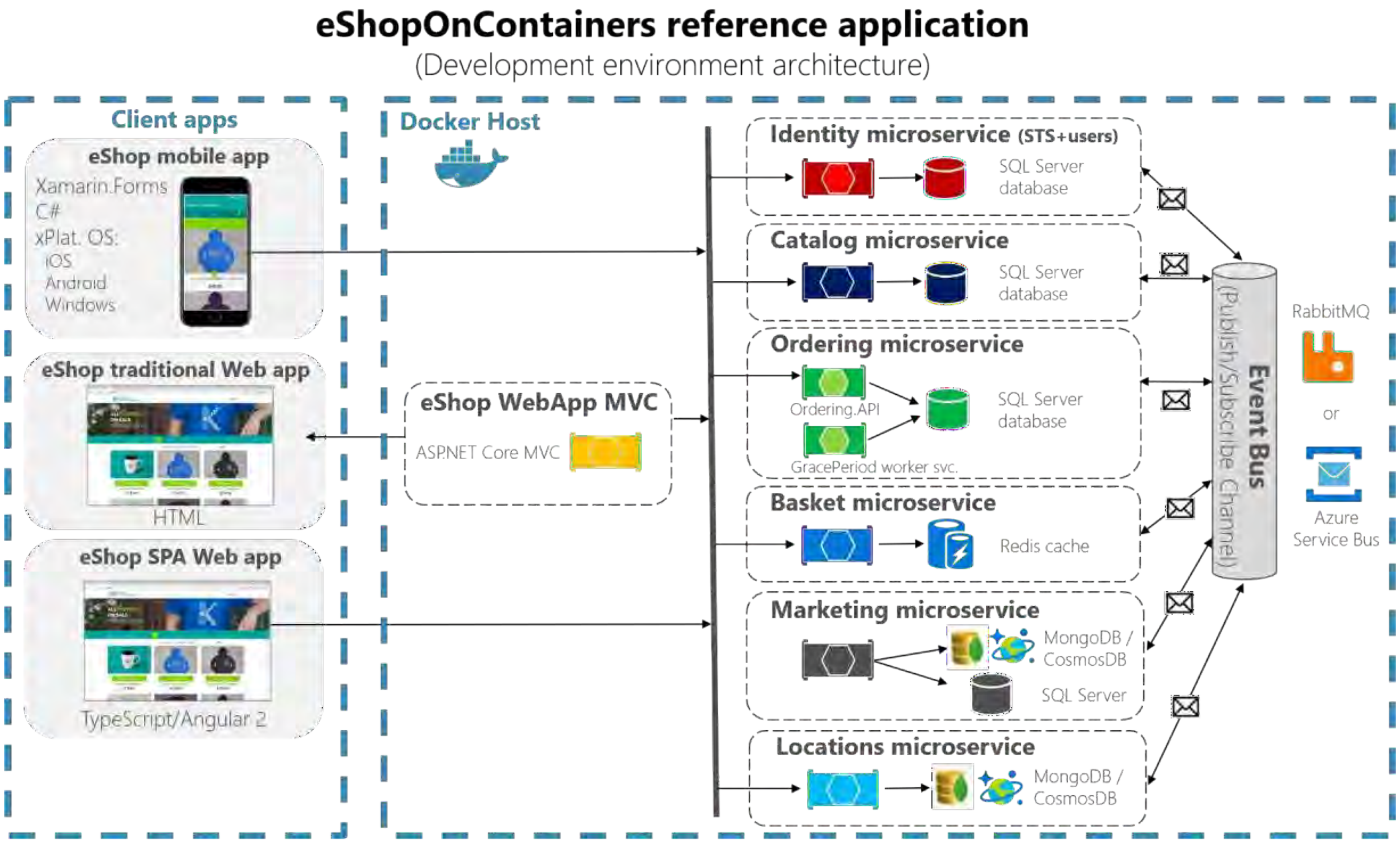
Microservices Architecture



- Distributed system
- Multiple machines
- Each service is a process
- Lightweight protocols

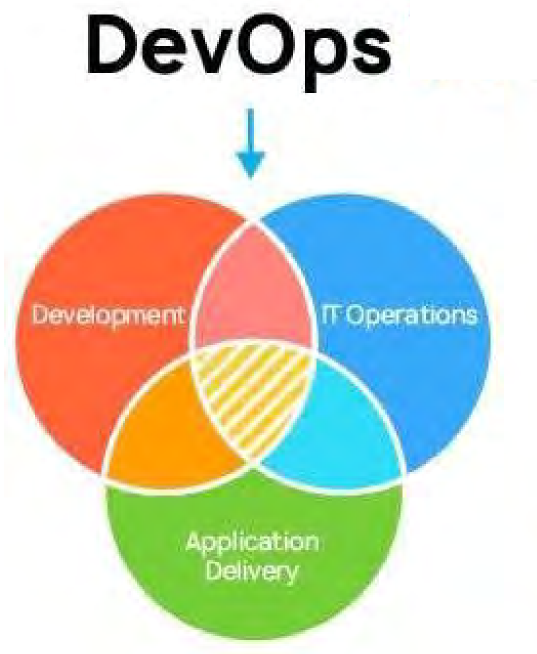


Microservices Architecture



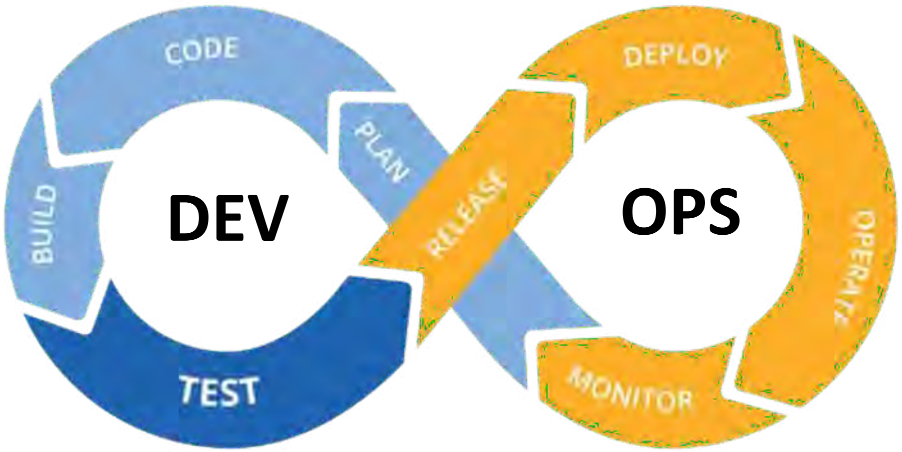
The Main Objectives of DevOps Adoption

- Faster delivery to market
- Reduction in errors and mistakes
- Cost efficiency
- Early error detection
- Rapid automation
- Association in the team
- Great feedback loop
- Etc



DevSecOps.

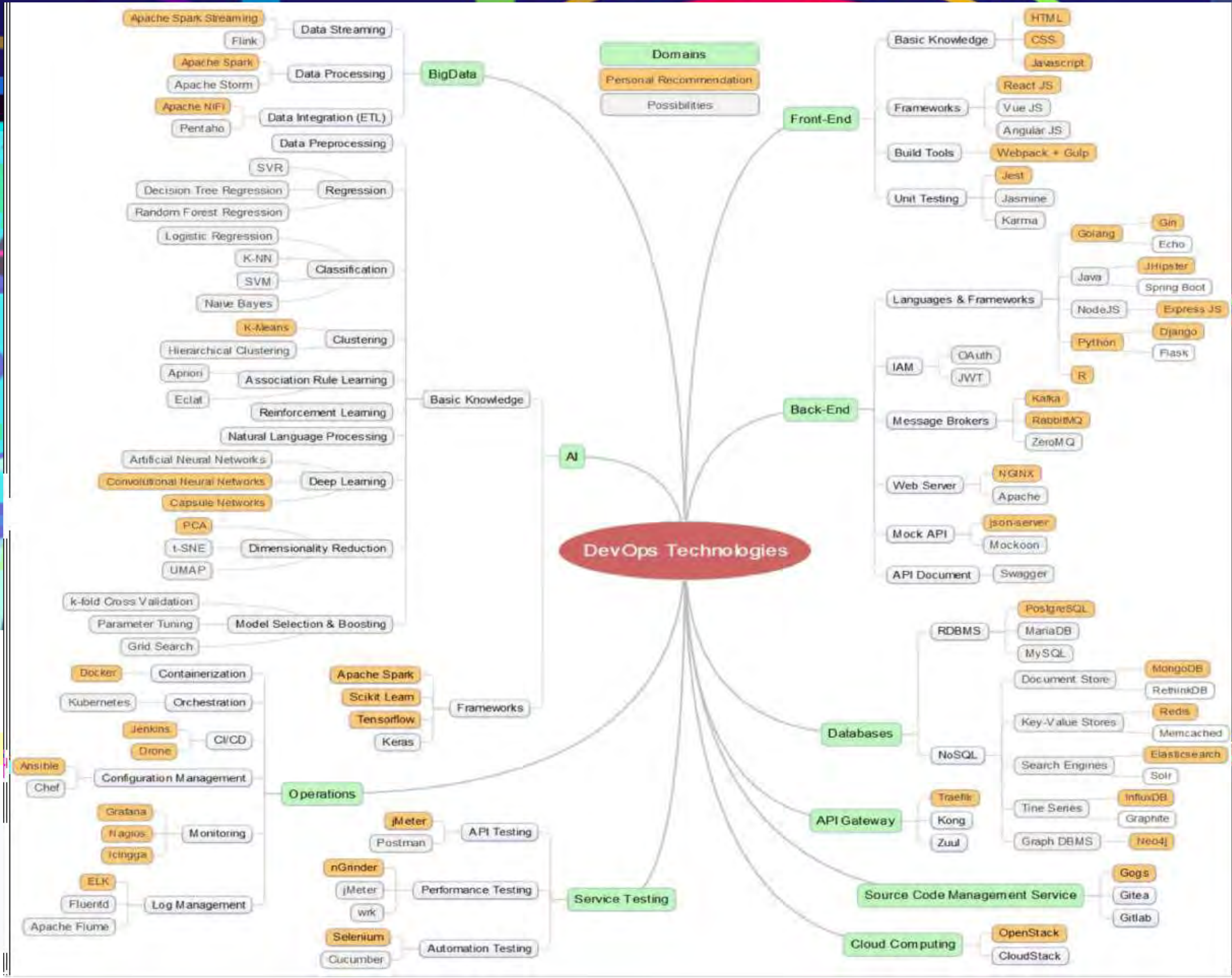
DevOps Lifecycle Management



DevOps roadmap requires time, effort, and adoption of DevOps strategy in every application or software lifecycle.

This becomes the DevOps lifecycle management.

DevOps Technologies Component



Application Technology Architecture

UI/UX dan Front End

1 **Framework**

Web Application Framework include Login Dialog, Menu, Landing Page

2 **Responsive Web/Mobile**

Responsive Web for Desktop and Mobile

3 **Form Template**

Web Template of Master, Transaction, Tab

4 **Data Search & Filter**

Predefine Search and Filter component

5 **Reporting**

Reporting with Jasper report engine

6 **Dashboard**

Components Template of Dashboard UI

Application Technology Architecture

Standard Web Service Integration

7 **Integration**

Web Service Framework with service token and session management

8 **Security**

Authentication, Access Management, User Role Management, Authentication,

9 **High Availability**

Message queue to increase availability of system

10 **High Availability**

Increase capacity with load balancing capability

Application Technology Architecture

Infrastructure Platform

PostgreSQL

Microsoft .NET

Java

RED HAT OPENSHIFT Container Platform

11 Not Only SQL

No SQLDB

Support no SQL language for elastic search

13 **OS Independent**

Java runtime for flexible and independent operating system

14 **Container**

Containerized docker to simplify and effective infrastructure

15 **Container**

Openshift platform to enable auto scaleup/down

Application Technology Architecture

Devops Platform

Jenkins

Nexus

GitLab

16 **Auto Deployment**

Automatic compile and deploy to the target container

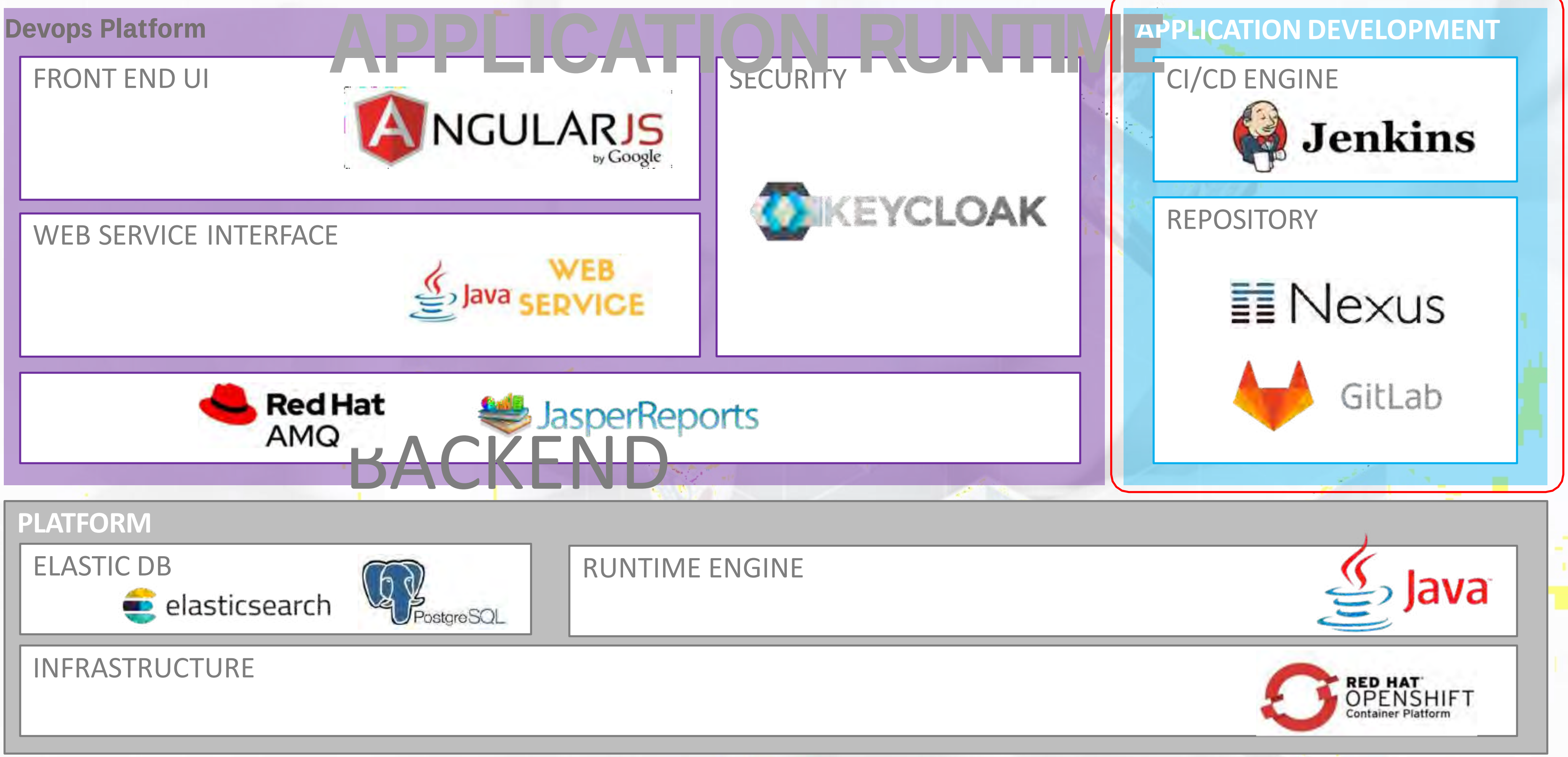
17 **Artifact Repository**

Artifact Repository for release management and quick rollback remediation

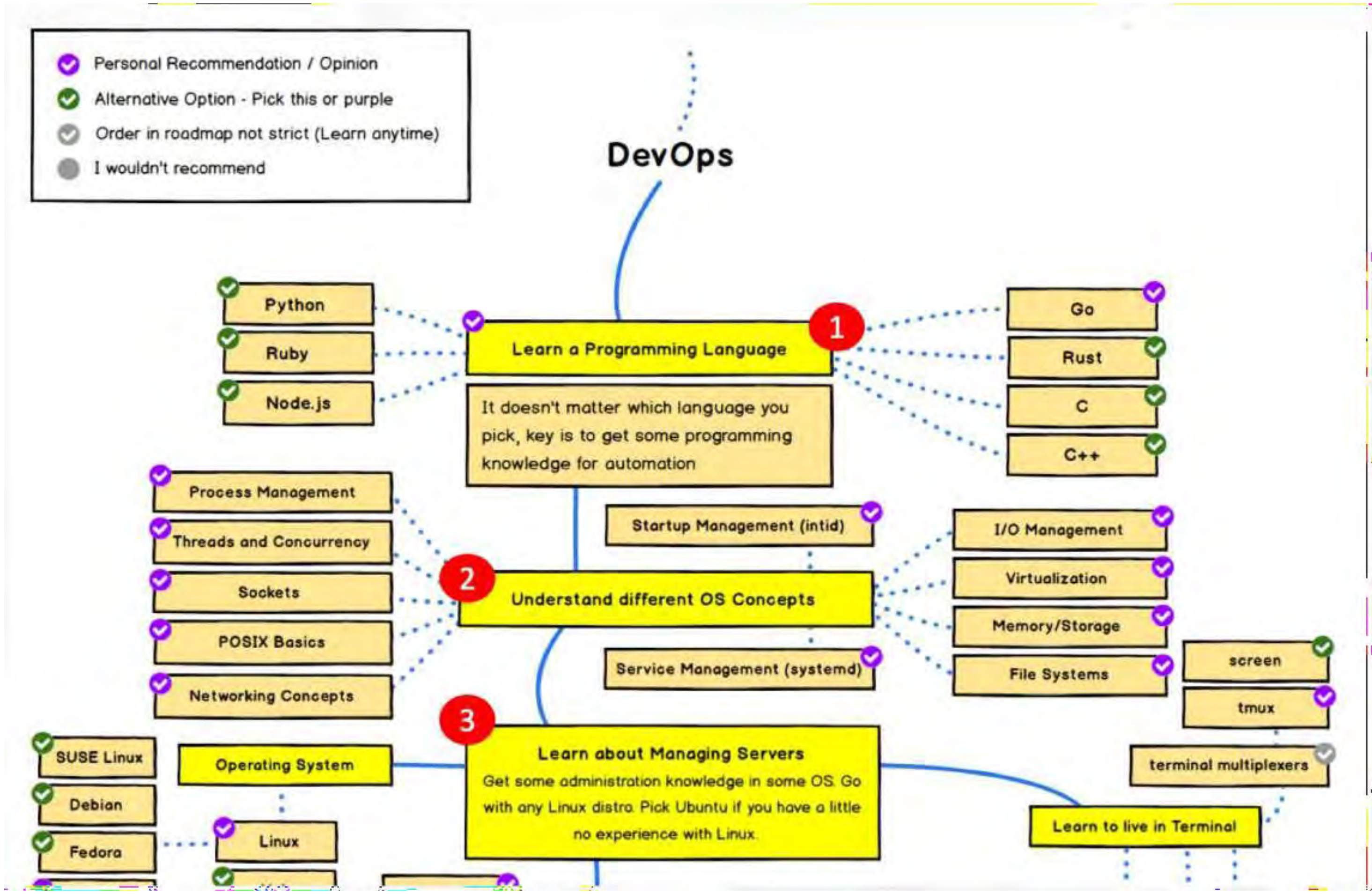
18 **Source Repository**

Integrated Source Repository

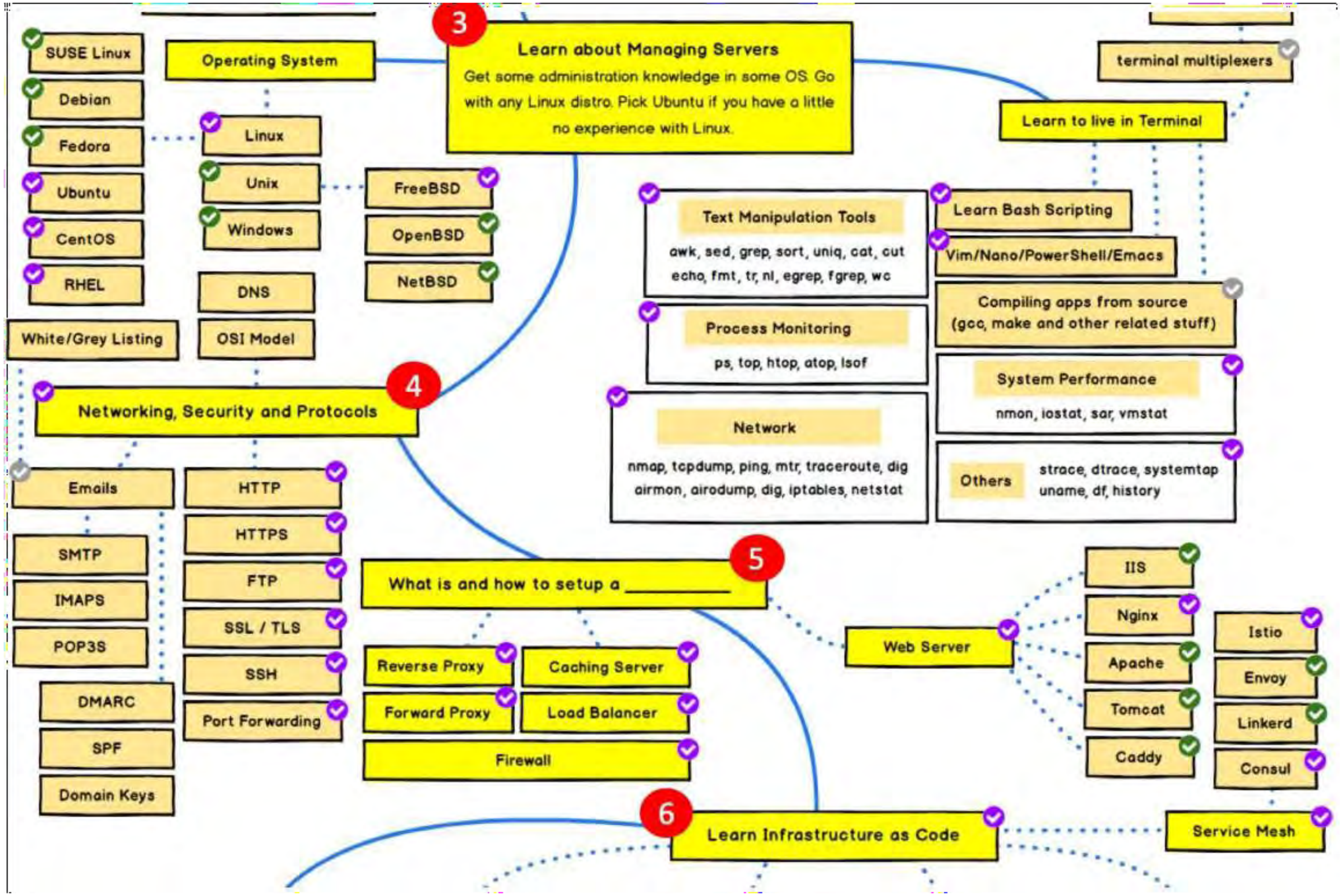
Application Technology Architecture



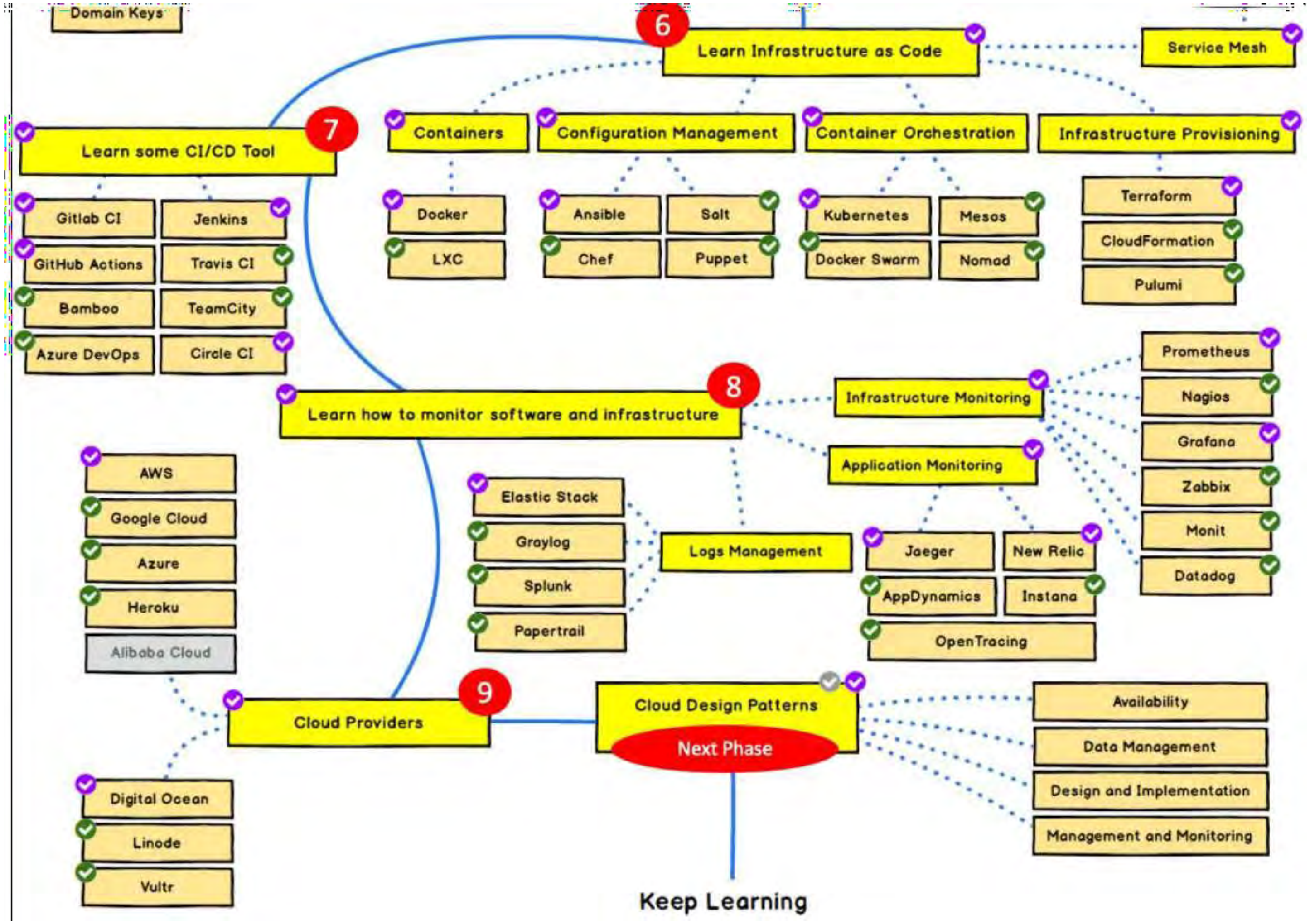
Devops Learning Path 1



Devops Learning Path 2

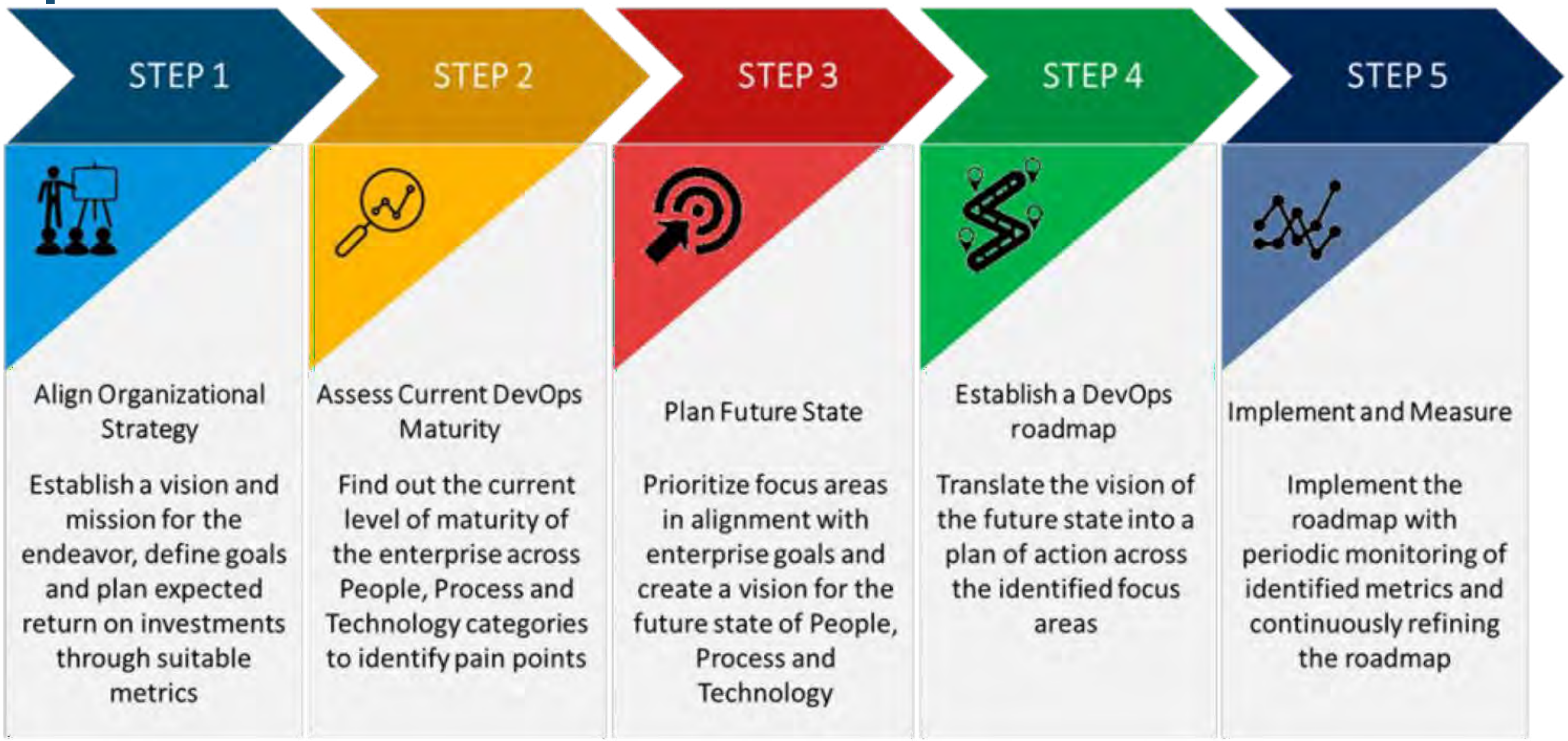


Devops Learning Path 3



Devops Adoption

Roadmap Devops Adoption



Maturity Assessment

Area:

1. Technology
2. Process
3. People
4. Culture

Area	ID	Initial level (1.)	Repeatable level (2.)	Defined level (3.)	Managed level (4.)	Optimized level (5.)
TECHNOLOGY	T1	Environments are provisioned manually	All environment configurations are externalized and versioned	Virtualization used if applicable	All environments managed effectively	Environment provisioning fully automated
	T2	Manual tests or minimal automation	Functional test automation	Triggered automated tests	Smoked tests and dashboard shared with Operational team	Chaos Monkey
	T3	Data migration un-versioned and performed manually	Changes to DB done with automated scripts versioned with application	DB changes performed automatically as part of deployment process	DB upgrades and rollbacks tested with every deployment	Feedback from DB performance after each release
	T4	Manual deployment	Build automation	Non-production deployment automation	Production deployment automation	Op.t. and Dev.t. regularly collaborate to manage risks and reduce cycle time
	T5	Manual processes for building software/ No artifact versioning	Regular automated build and testing. Any builds can be recreated from source	Automated build and test cycle every time a change is committed	Build metrics gathered, made visible and take into account	Continuous work on process improvement, better visibility, faster feedback
	T6	No collaboration tools	Project planning tool	Team/ toolset integration	Knowledge management tool	-
	T7	No software configuration management (SCM)	Standardized SCM	Configuration is delivered together with code	Self-healing tools	-
	T8	No or minimal monitoring	Core monitoring	Integrated monitoring	Analytics/ Intelligence	-
	T9	No tools or minimal tool usage for issue tracking	All issue and bug reports are tracked	Issue reporting automatization and monitoring	Activities based on received feedback and data	Continuous delivery process
	PR1	Inconsistent delivery	Scheduled delivery	Automated delivery	Frequent delivery	Development process integrated with Six

Sample High Level Roadmap

Stage 3 – Optimized

Stage 2 – Standardized

Stage 1 – Quick Win

Proof-of-concept implementations

- DevOps **implementation planned**
- **Tools & processes** chosen
- **3-4 teams** chosen
- **Teams** mentored
- **Implementation is done** and lessons learned

Org-wide Adoption

- **Mentoring sessions** across the organization
- DevOps **rolled out** for **all the teams**
- DevOps implementations **governed**
- Sustained DevOps implementation **across different teams**
- **Tracking & reporting**

Optimized DevOps

- Lessons learned using **continuous feedback** is incorporated **back to improve** the DevOps implementations
- The appropriate **mix of tools & frameworks** used for **optimized outcomes**
- **Development processes updated** to optimize DevOps outcomes

Quick-win

The **first phase of transformation is very important** step to increase the **winning spirit** of **IT Team, Business Team and Management**.

Thus the selection of step will be consider some important aspects of:

Basic function for Startup Operation.

In this objective will be basic feature of **Adoption of Devops** as quick-win piloting project.

The project will be executed **quick-win within 1-3 months** with limited focus on customer's application function

and by **6-12 months** development of Comprehensive Devops

"We commit to delivering the best value solutions
for our clients"